
Somatic Variant Callers Agree on Clonal Variants but Not on Subclonal or Repeat-Region Variants: A VAF-, Type- and Context-Stratified Analysis

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Abstract

Cancer genomics pipelines often report a somatic mutation only when several variant callers agree on it, so it matters which variants a consensus loses. We ask whether callers agree on more than 80% of variants above 20% variant allele fraction (VAF) and on fewer than 50% below 5% VAF, and whether indels and variants in low-complexity (LC) regions disagree more at every VAF. We measure the fraction of variants called by all callers, stratified jointly by VAF, variant type and LC context, in three settings. These are five sequencing runs of the SEQC2 HCC1395 reference tumor (five SNV and three indel callers), the SEQC2 tumor-purity dilution series, and 10,074 TCGA MC3 exomes. On truth-set variants, unanimity is 88–96% for SNVs and 86–91% for indels at $\geq 20\%$ VAF, and $\leq 0.3\%$ for SNVs and 19–37% for indels below 5% VAF. Diluting the same 16,636 clonal SNVs from 100% to 5% purity lowers unanimity from 93% to 0.6%, which shows that VAF itself causes the drop. On the union of all calls, however, only 29–56% of high-VAF SNV calls are unanimous. LC regions lower unanimity in every VAF bin where agreement is above its floor (adjusted odds ratios 0.02–0.72). Indels disagree more than SNVs at $\geq 20\%$ VAF, but at lower VAF the ranking depends on which callers are compared. All-caller consensus is thus a precise filter for clonal variants, but it discards nearly all subclonal variants and much of the true signal in repeats.

1 Introduction

A somatic mutation that enters a cancer study has usually passed through a vote. Large efforts such as TCGA MC3 [Ellrott et al., 2018], the ICGC benchmark [Alioto et al., 2015] and the SEQC2 reference project [Fang et al., 2021, Xiao et al., 2021] combine several somatic variant callers, and many pipelines keep only mutations that all or most callers report [Goode et al., 2013, Fang et al., 2015, Wang et al., 2020]. Whether a mutation is reported therefore depends directly on whether callers agree on it. Our main question is simple: **when several callers analyse the same tumor, how often do they all agree, and on which variants do they fail to agree?**

The answer matters most for the variants that are hardest to call. Subclonal mutations have low variant allele fraction (VAF) and can drive therapy resistance. Indels in homopolymers and tandem repeats drive the microsatellite-instability phenotype. If callers routinely disagree on these variants, a consensus filter removes them without warning. A common working belief is that callers agree on most high-VAF variants, disagree on most low-VAF variants, and disagree more on indels and on low-complexity (LC) sequence at any VAF. We state this belief as four testable hypotheses. **H1:** all callers agree on more than 80% of variants with VAF $\geq 20\%$. **H2:** they agree on fewer than 50% of variants with VAF $< 5\%$. **H3:** indels disagree more than SNVs in every VAF bin. **H4:** variants in LC regions disagree more than other variants in every VAF bin.

Prior work does not settle these hypotheses. Only one study reports caller concordance stratified by VAF, and it compares two callers on SNVs only [Chen et al., 2020]. Many benchmarks show that indels are harder [Alioto et al., 2015, Krøigård et al., 2016, Guille et al., 2025], but none tests whether the indel penalty persists within each VAF stratum. Evidence for an LC penalty comes mostly from germline benchmarks [Dwarshuis et al., 2024] or from somatic truth sets that simply exclude such regions [Jones et al., 2021]. Finally, published agreement numbers use different denominators. Goode et al. [2013] report 46% agreement on all calls at $\text{VAF} \geq 20\%$, while Wang et al. [2013] report that 82% of validated SNVs are called by all five callers. The 80% and 50% thresholds are ambiguous until the denominator is fixed.

We address these gaps with a joint $\text{VAF} \times \text{variant type} \times \text{LC}$ analysis of multi-caller agreement. We use released per-caller outputs for the SEQC2 HCC1395 tumor: five sequencing runs with five SNV and three indel callers, plus a six-caller raw-VCF replicate. We report agreement on two denominators: the TRUTH set of validated variants, and the UNION of everything any caller calls. To separate the effect of VAF from site properties that co-vary with VAF, we use the SEQC2 tumor-purity dilution series, where the *same* variants appear at lower VAF. We then replicate the analysis on 10,074 TCGA exomes from MC3.

The results support the hypothesis only in part, and the denominator decides H1. On truth variants, callers agree on 88–96% of SNVs and 86–91% of indels at $\text{VAF} \geq 20\%$ in the four standard-depth runs, and on at most 0.3% of SNVs below 5% VAF. Diluting 16,636 clonal SNVs from 100% to 5% purity lowers unanimity from 0.928 to 0.006 (McNemar $p < 10^{-300}$), so VAF is the cause. On the union of calls, only 29–56% of high-VAF SNV calls are unanimous, and 63% of SNV calls come from a single caller. LC regions lower unanimity in every bin where agreement is above zero (adjusted odds ratio 0.02–0.72), with no bin reversed. Indels disagree more than SNVs at $\text{VAF} \geq 20\%$ in every dataset. At lower VAF the ranking reverses when each type uses its own caller panel, because the SNV panel contains SomaticSniper, which makes almost no true calls below 10% VAF.

In summary, our contributions are:

- We test four pre-registered hypotheses on somatic caller agreement with a joint $\text{VAF} \times \text{type} \times \text{LC}$ stratification, two explicit denominators and within-bin contrasts, across five SEQC2 runs and a six-caller replicate.
- We use a tumor-purity dilution series as a within-variant test and show that VAF alone drives the collapse of agreement.
- We show that the LC penalty holds in every measurable VAF bin, whereas the indel penalty below 20% VAF depends on the caller panel, and we trace low-VAF disagreement to the sensitivity of the weakest caller.
- We replicate the main findings on 10,074 TCGA MC3 exomes and derive practical guidance for consensus calling.

Section 2 reviews prior work. Section 3 describes data, denominators and statistics. Section 4 reports results for each hypothesis. Section 5 discusses implications and limitations, and section 6 concludes.

2 Related Work

Benchmarks of somatic callers. Somatic callers compare tumor and matched-normal reads using different models. Some are likelihood-based (MuTect, MuTect2, MuSE, Strelka) [Cibulskis et al., 2013, Benjamin et al., 2019, Fan et al., 2016]. Some use heuristics (VarScan2, VarDict, SomaticSniper). Some use local assembly (MuTect2, Lancet, TNscope), and some use deep learning [Sahraeian et al., 2019]. Benchmarks have found low raw overlap between callers again and again. In the ICGC benchmark, only 205 SNVs and a single indel were called by all pipelines [Alioto et al., 2015]. Krøigård et al. [2016] found 472 SNVs called by all eight SNV callers but 22,032 called by only one. Cai et al. [2016] found that only about 21% of SNVs were called by at least two callers. The DREAM challenge used synthetic tumors to show that ensembles help [Ewing et al., 2015]. de Schaetzen van Brien et al. [2020] found that four callers shared only 8 calls. These studies mostly report precision and recall against a truth set, or overall overlap counts. Unlike them, we measure agreement jointly by VAF, variant type and sequence context.

Agreement as a function of VAF. Caller sensitivity drops at low VAF, and it drops at different rates for different callers. At 5% VAF and $30\times$ depth, MuTect reached 16% sensitivity and Strelka reached

4.6% [Cibulskis et al., 2013]. Below 5% VAF, true-positive rates ranged from 0.22 to 0.70 [Sergi et al., 2024], and VarScan and SomaticSniper failed at $\leq 10\%$ VAF [Wang et al., 2020]. de Schaetzen van Brien et al. [2020] reported F1 of 0.82–0.95 for clonal variants but 0–0.72 for subclonal variants. The only direct VAF-stratified concordance comes from Chen et al. [2020]: Strelka2 and MuTect2 agreed on more than 90% of SNVs at $\geq 10\%$ VAF, 75–89% at 5% and 20–41% at 1%. Goode et al. [2013] showed that restricting to VAF $\geq 20\%$ raised three-caller agreement only to 46% of all calls. Wang et al. [2013], by contrast, found that 82% of validated SNVs were called by all five callers. Building on these studies, we use five to six callers, analyse both SNVs and indels, report both denominators, and add a dilution design that changes VAF while holding the variants fixed.

Indels and low-complexity sequence. Indels are harder to call than SNVs. The best indel F1 in the ICGC benchmark was 0.65, compared with 0.79 for SNVs, and 83% of true indels lay in tandem repeats [Alioto et al., 2015]. Guille et al. [2025] found that 11 of 18 callers reached F1 above 0.8 for SNVs, but only 2 of 15 did so for indels. Benjamin et al. [2019] report indel sensitivity 10–30 points below SNV sensitivity at matched allele fraction. In germline benchmarks, the GIAB stratifications show lower precision and recall in homopolymers and tandem repeats [Dwarshuis et al., 2024]. Somatic reference sets often exclude these regions because results there do not reproduce [Jones et al., 2021]. Because most indels sit in repeats, type and context effects are confounded in these studies, and neither effect has been tested within VAF strata. We separate them by comparing LC with non-LC variants within each type and each VAF bin, adjusting for mappability and segmental duplications.

Reference data. The SEQC2 consortium sequenced the HCC1395 breast-cancer cell line and its matched normal at several centers, called variants with several pipelines, and released a high-confidence truth set together with a purity and depth series [Fang et al., 2021, Xiao et al., 2021, Zhao et al., 2021]. TCGA MC3 released pan-cancer calls from seven callers, listing for each variant the callers that reported it [Ellrott et al., 2018]. The SomaticSeq ensemble tables [Fang et al., 2015] record per-caller flags for every candidate site. We are the first, to our knowledge, to use these resources jointly to test stratified agreement hypotheses.

3 Methods

3.1 Problem formulation

Consider a set of K callers $\mathcal{C} = \{c_1, \dots, c_K\}$ applied to the same tumor–normal pair, and a set of sites $\mathcal{S}$. For site $s \in \mathcal{S}$, let $n(s) = \sum_{k=1}^K \mathbf{1}[c_k \text{ calls } s]$. Our primary metric for a stratum $\mathcal{S}' \subseteq \mathcal{S}$ is the **all-agree fraction**

$$A(\mathcal{S}') = \frac{1}{|\mathcal{S}'|} \sum_{s \in \mathcal{S}'} \mathbf{1}[n(s) = K], \quad (1)$$

and disagreement is $1 - A$. We stratify by VAF bin ($< 5\%$, $5\text{--}10\%$, $10\text{--}20\%$, $\geq 20\%$), variant type (SNV or indel) and LC status. Secondary metrics are the strict-majority fraction ($n > K/2$), the mean fraction of callers n/K , the singleton fraction ($n = 1$), the mean pairwise Jaccard index, Fleiss’ κ , and per-caller call rate, which equals recall on the truth set. Following the pre-registered plan, the hypotheses are H1: $A > 0.80$ at VAF $\geq 20\%$; H2: $A < 0.50$ at VAF $< 5\%$; H3: $1 - A$ is higher for indels than SNVs in every bin; and H4: $1 - A$ is higher in LC than non-LC regions in every bin, for each type.

3.2 Data

Table 1 summarises the four experiments.

D1: SEQC2 ensemble tables. We use the SomaticSeq [Fang et al., 2015] ensemble tables for five whole-genome runs of HCC1395 [Fang et al., 2021, Zhao et al., 2021]. Four are standard-depth runs (IL_T_1, NV_T_1, FD_T_1, NS_T_1; about $30\text{--}60\times$), and NS_380X combines nine NovaSeq runs ($380\times$). The SNV panel is MuTect2, SomaticSniper, VarDict, MuSE and Strelka ($K = 5$). The indel panel is MuTect2, VarDict and Strelka ($K = 3$), because SEQC2 provides no SomaticSniper or MuSE indel calls. We use PASS-like call flags. A call means MuTect2 flag = 1, VarDict flag = 1 (PASS and Somatic), MuSE Tier 1, Strelka flag = 1, and SomaticSniper SS = 2 with $\text{SSC} \geq 40$. A lenient variant also counts VarDict flag 0.5, any MuSE tier and any SomaticSniper SS = 2.

Exp.	Data	SNV callers (K)	Indel callers (K)	Scale
D1	SEQC2 HCC1395 WGS, SomaticSeq tables	MuTect2, SomaticSniper, VarDict, MuSE, Strelka (5)	MuTect2, VarDict, Strelka (3)	5 runs, 30–380×
D1B	SEQC2 IL_T_1 raw PASS VCFs	D1 callers + TNscope (6)	D1 callers + TNscope (4)	1 run
D2	SEQC2 purity series, 100× WGS	TNscope, Lancet, MuTect2, Strelka, SomaticSniper (5)	TNscope, Lancet, MuTect2, Strelka (4)	6 purities
D3	TCGA MC3 public MAF, WES	MuTect, MuSE, VarScan2, SomaticSniper, RADIA (5)	Indelocator, VarScan2, Pindel (3)	10,074 / 8,630 tumors

Table 1: Experiments. D1, D1B and D2 share one tumor (HCC1395) and one truth set (39,447 SNVs, 1,625 indels). D3 has no truth set, and its public MAF contains only variants called by at least two callers.

D1B: raw VCFs. To check that results do not depend on SomaticSeq flag semantics, we parse the raw PASS calls of IL_T_1 and add TNscope. This gives six SNV callers and four indel callers.

D2: purity dilution series. SEQC2 diluted the tumor into the normal at 100, 75, 50, 20, 10 and 5% purity and sequenced each mixture at 100×. The released PASS calls come from TNscope, Lancet, MuTect2 and Strelka, plus SomaticSniper for SNVs. For a truth variant with pooled allele fraction TVAF, the expected VAF at purity π is $\text{TVAF} \times \pi$. The dilution changes VAF while the variants stay fixed, so it tests the effect of VAF within variants.

D3: TCGA MC3. We use the public MC3 MAF (GRCh37, v0.2.8) [Ellrott et al., 2018]. The SNV callers are MuTect, MuSE, VarScan2, SomaticSniper and RADIA. The indel callers are Indelocator, VarScan2-indel and Pindel. A caller counts as calling a variant if it appears in the CENTERS field; a trailing * also counts, consistent with NCALLERS. We keep only tumors in which all callers of a type were active, so K is constant: 10,074 tumors for SNVs and 8,630 for indels. The public MAF contains only variants called by at least two callers, so MC3 agreement is conditional on $n \geq 2$ and is an upper bound.

Truth set and annotations. The truth set is SEQC2 v1.2.1: 39,447 SNVs and 1,625 indels in high-confidence (HC) regions on chr1–22 and X. We define LC regions with the GIAB v3.3 stratum A11TandemRepeatsandHomopolymers_slop5 [Dwarshuis et al., 2024]. We also annotate GIAB low-mappability regions, segmental duplications and SEQC2 HC regions, computing overlaps with bedtools 2.31 on the reference-allele span. Indels from the purity series and raw VCFs are left-normalised against hg38 before exact matching.

3.3 Denominators

Prior numbers differ mainly in what they count [Goode et al., 2013, Wang et al., 2013], so we fix the denominator explicitly.

- **TRUTH:** all truth variants, including those no caller detects. VAF is the pooled TVAF from about 63 SEQC2 replicates. It does not depend on the callers or on detection. In D2 we use $\text{TVAF} \times \pi$. As a robustness check, we also use the VAF observed in each run.
- **UNION:** every site called by at least one caller. VAF comes from tumor read counts in D1 and from the median caller-reported VAF in D2.
- **UNION-HC:** UNION restricted to SEQC2 HC regions.

TRUTH measures whether callers agree on real mutations, so it isolates sensitivity. UNION measures whether callers agree on what they report, so it also reflects each caller’s private false positives.

3.4 Statistical analysis

H1 and H2. We use one-sided exact binomial tests of A against 0.80 (H1) and 0.50 (H2), with Holm correction within each experiment.

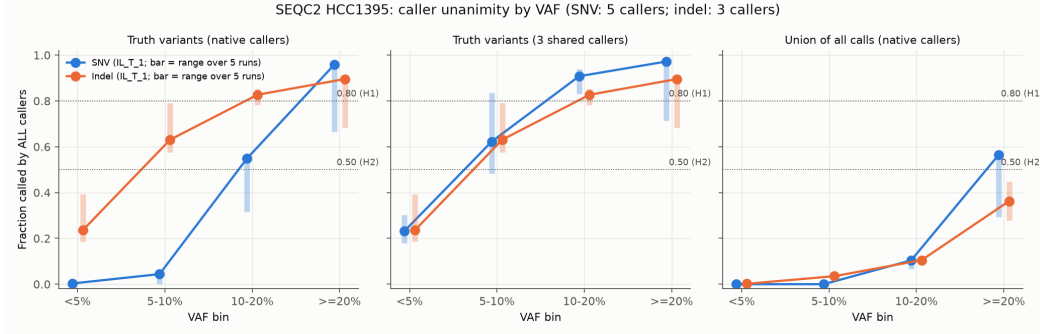


Figure 1: Fraction of sites called by all callers in SEQC2 D1, by VAF bin. Points show IL_T_1, and bars show the range over the five runs. Dotted lines mark the H1 (0.80) and H2 (0.50) thresholds. (Left) Truth variants with native panels (5 SNV callers, 3 indel callers). (Center) Truth variants with the three shared callers. (Right) Union of all calls. Truth variants pass 0.80 at $\geq 20\%$ VAF, except in the $380\times$ run, but the call union never does.

H3 and H4. Within each VAF bin, we compute the risk difference in disagreement with a Newcombe 95% confidence interval (CI). We test it with a χ^2 or Fisher test and apply Holm correction. Cells with $n < 20$ are excluded. A context counts as *supported (all bins)* if the CI lies above 0 and $p_{\text{Holm}} < 0.05$ in every tested bin. It counts as *partial* if it is supported in some bins and significantly reversed in none, and as *mixed* if it is significantly supported in some bins and significantly reversed in others.

Regression. For each type and denominator we fit the logistic model

$$\text{logit } P(n = K) = \text{VAF_bin} \times \text{LC} + \log(1 + \text{depth}) + \text{lowmap} + \text{segdup}. \quad (2)$$

A pooled model on the shared callers adds $\text{VAF_bin} \times \text{is_indel}$. We test the interactions with likelihood-ratio tests (LRTs). A significant interaction means that the size of the penalty changes with VAF.

Caller-set controls. SNV and indel panels differ in size and membership, so a naive H3 comparison is confounded by the caller panel. We therefore also compare types on *shared* panels: MuTect2, VarDict and Strelka in D1 (shared-3), and the same three callers plus TNScope in D1B (shared-4). In D2, shared-4 is TNScope, Lancet, MuTect2 and Strelka. A leave-one-caller-out analysis in D1B shows which caller drives disagreement.

Paired and clustered designs. In D2, we compare the same clonal variants ($\text{TVAf} \geq 0.4$) at each purity with 100% purity using McNemar’s test. In D3, variants within a tumor are not independent, so we use a tumor-level cluster bootstrap (200 replicates, seed 42) and also report per-tumor macro averages.

Implementation. The analysis uses Python 3.12 with pandas, numpy, scipy, statsmodels and pysam, plus bedtools 2.31. It runs on a 32-thread CPU in about 45 minutes, most of it spent fitting the D1 regression models. The only stochastic step is the D3 bootstrap. Re-running it reproduced byte-identical outputs.

4 Results

We report results for each hypothesis in turn. We first ask **how agreement depends on VAF** (H1, H2; section 4.1), then **whether LC regions disagree more** (H4; section 4.2), then **whether indels disagree more** (H3; section 4.3). We end with a comparison to published numbers (section 4.4).

4.1 Agreement collapses with VAF, but H1 depends on the denominator

On truth variants, H1 and H2 hold. Table 2 and figure 1 show a steep rise in agreement with VAF. In the four standard-depth runs, all callers agree on 0.877–0.964 of truth SNVs and 0.865–0.909 of truth indels at $\text{VAF} \geq 20\%$. These values exceed 0.80 (one-sided $p_{\text{Holm}} < 0.001$). Below 5% VAF,

Denominator	Type (K)	VAF bin			
		<5%	5–10%	10–20%	$\geq 20\%$
TRUTH	SNV (5)	0.003 [0–0.003]	0.044 [0.001–0.044]	0.549 [0.317–0.549]	0.959 [0.666–0.964]
	Indel (3)	0.235 [0.186–0.392]	0.631 [0.575–0.791]	0.827 [0.782–0.853]	0.896 [0.684–0.909]
UNION	SNV (5)	0.000	0.000	0.104 [0.066–0.104]	0.564 [0.293–0.564]
	Indel (3)	0.002 [0–0.007]	0.035 [0.022–0.052]	0.105 [0.081–0.121]	0.363 [0.278–0.448]

Table 2: All-agree fraction in SEQC2 D1 (strict flags, native callers). Values are for IL_T_1, with the range over all five runs in brackets. The TRUTH set in IL_T_1 has 1,292 / 3,853 / 6,271 / 28,030 SNVs and 102 / 287 / 422 / 814 indels per bin. The UNION set has 83,570 / 74,665 / 37,199 / 156,082 SNV sites and 17,556 / 6,587 / 5,939 / 19,452 indel sites. Bold values exceed the H1 threshold of 0.80; the lower end of their ranges comes from the 380 $\times$ run. H1 holds only on the TRUTH denominator.

Type	<5%	5–10%	10–20%	$\geq 20\%$
SNV	0.174	0.632	0.864	0.925
Indel	0.100	0.472	0.672	0.730

(a) All-agree by expected VAF ($\text{TVAF} \times \pi$), pooled over six purities, shared-4 callers. All 95% CIs are within ± 0.02 .

Purity	SNV ($K=5$)	Indel ($K=4$)
100%	0.928	0.742
50%	0.925	0.685
20%	0.494 ↓	0.324 ↓
10%	0.082 ↓	0.088 ↓
5%	0.006 ↓	0.024 ↓

(b) The same clonal variants ($\text{TVAF} \geq 0.4$; 16,636 SNVs, 330 indels) at each purity.

Table 3: D2 purity dilution series. Lowering purity lowers VAF while keeping the variants fixed. For SNVs, 15,350 clonal variants lost unanimity between 100% and 5% purity and only one gained it (McNemar $p < 10^{-300}$).

unanimity is at most 0.003 for SNVs in every run. For indels it is 0.19–0.37 and significant in four of five runs; in NS_380X it is 0.39 (95% CI 0.30–0.49), which is not significant. Majority agreement is much higher. For IL_T_1 truth SNVs it is 0.235, 0.647, 0.941 and 0.996 across the four bins.

More depth does not guarantee more agreement. H1 fails in the 380 $\times$ run (SNVs 0.666, indels 0.684). The cause is two callers. At $\text{VAF} \geq 20\%$, MuTect2 recall falls to 0.72 and MuSE recall to 0.68 at this depth, while the other callers stay at 0.998.

On the call union, H1 fails everywhere. Only 0.293–0.564 of union SNV sites and 0.278–0.448 of union indel sites at $\text{VAF} \geq 20\%$ are unanimous. This holds for raw PASS calls, for calls restricted to HC regions, and for the lenient flag definition. In IL_T_1, only 18% of the $\geq 20\%$ SNV union are truth sites, and 28% are single-caller calls. Across all bins, 63% of union SNVs and 70% of union indels are singletons. H2 holds everywhere on the union ($A \leq 0.007$).

Six raw callers give the same picture. With raw PASS VCFs and TNscope added (D1B), truth unanimity at $\geq 20\%$ VAF is 0.933 for SNVs ($K = 6$) and 0.857 for indels ($K = 4$). Below 5% it is 0.002 and 0.088. Union unanimity at $\geq 20\%$ is 0.548 and 0.240.

Dilution shows that VAF itself causes the collapse. Observed VAF co-varies with copy number, mappability and sequence context, so the pattern above could reflect site properties rather than VAF. The purity series rules this out (table 3, figure 2). The same 16,636 clonal SNVs are unanimous in 0.928 of cases at 100% purity, 0.494 at 20%, 0.082 at 10% and 0.006 at 5%. Between 100% and 5% purity, 15,350 variants lost unanimity and one gained it (McNemar $p < 10^{-300}$). Clonal indels fall from 0.742 to 0.024. Because the variants are identical, site properties cannot explain the drop; it follows the lower VAF.

MC3 replicates the pattern. Across 10,074 TCGA exomes, all five SNV callers agree on 0.841 of variants at $\text{VAF} \geq 20\%$ (cluster-bootstrap CI 0.838–0.844). They agree on 0.285 at 10–20% and on none below 10% (figure 3). The per-tumor median at $\geq 20\%$ is 0.844. The zeros are structural. MC3’s SomaticSniper made no calls below 10% VAF, and VarScan2 made none below 5%, so every public SNV below 5% was called by exactly MuTect and MuSE. For indels, all three callers agree on 0.638 at $\geq 20\%$, 0.643 at 10–20% and 0.213 at 5–10%. The public MAF has no indels below 5%.

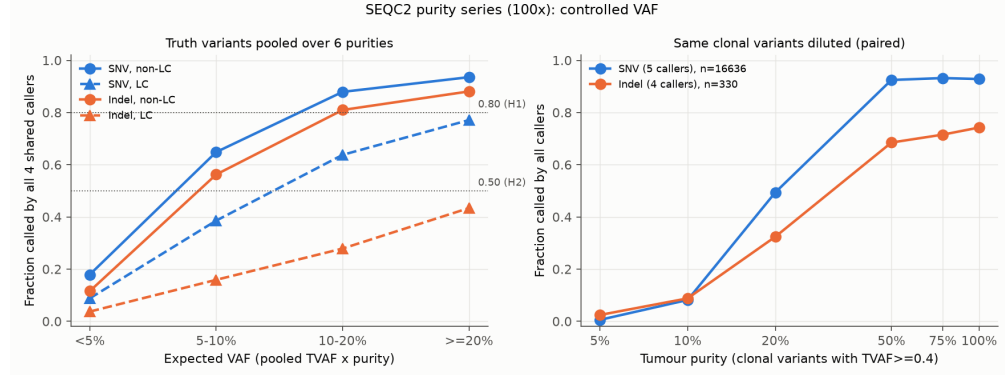


Figure 2: D2 purity dilution series (100 $\times$). (Left) Truth variants pooled over six purities, binned by expected VAF (TVAF $\times$ purity), with the four shared callers. LC variants (dashed) have lower agreement than non-LC variants in every bin, and LC indels have the lowest. (Right) The same clonal variants (TVAF ≥ 0.4) at decreasing purity. Unanimity stays near its maximum down to 50% purity and collapses below 20%.

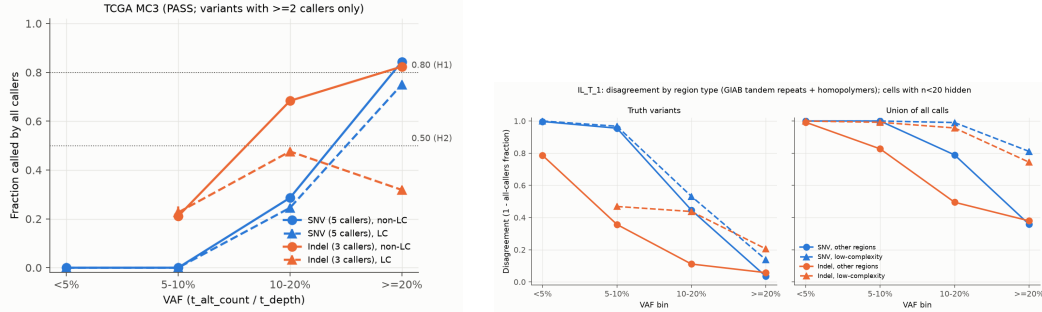


Figure 3: TCGA MC3 (PASS; variants with ≥ 2 callers). SNV unanimity is zero below 10% VAF because SomaticSniper makes no calls there. LC indels (dashed orange) fall far below non-LC indels at $\geq 20\%$ VAF.

Figure 4: Disagreement ($1 - A$) by region in IL_T_1 (native panels; cells with $n < 20$ hidden). (Left) Truth variants. (Right) Union of calls. Dashed LC lines lie above solid non-LC lines in every bin where disagreement is not saturated near 1.

4.2 Low-complexity regions disagree more wherever agreement is measurable

LC regions have lower agreement in every informative cell. Table 4 and figure 4 compare LC and non-LC sites. On the union at $\geq 20\%$ VAF, LC disagreement exceeds non-LC disagreement by 0.45 for SNVs and 0.36 for indels. Within HC regions, the gaps grow to 0.72 and 0.38. On truth variants at 10–20% VAF, the gaps are 0.09 for SNVs and 0.33 for indels. In the dilution series with shared-4 callers, LC truth variants are worse in every bin at both 100% purity (SNVs +0.09 to +0.23) and 20% purity. LC indels are worse in every tested bin at both purities (+0.10 to +0.45). In MC3 at $\geq 20\%$ VAF, the gap is 0.09 for SNVs and 0.51 for indels. LC sites are also common: they make up 60% of union SNVs and 88% of union indels. Among truth variants, which are restricted to HC regions, the shares are 6% and 24%.

No bin reverses, but many bins are at the floor. No bin in any dataset shows LC significantly better than non-LC. In D1, 39 contexts are supported in all bins, 50 are partial and 1 is mixed. D1B has 4 supported and 8 partial contexts, D2 has 24 supported, 44 partial and 4 not significant, and D3 has 3 partial and 1 not significant. The partial verdicts arise almost entirely from floor effects. Below 5–10% VAF, unanimity is near zero in both LC and non-LC regions, and some truth LC cells are small. H4 therefore holds wherever the metric can move, but “regardless of VAF” holds only trivially at the floor.

Denominator	Type	Region	<5%	5–10%	10–20%	≥20%
TRUTH	SNV	non-LC	0.231	0.626	0.916	0.977
		LC	0.240 [†]	0.567	0.776	0.896
	Indel	non-LC	0.213	0.643	0.889	0.942
		LC	0.385 [†]	0.531 [†]	0.562	0.795
UNION	SNV	non-LC	0.002	0.093	0.419	0.794
		LC	0.000	0.005	0.023	0.246
	Indel	non-LC	0.008	0.173	0.505	0.620
		LC	0.000	0.008	0.043	0.256

Table 4: All-agree fraction in LC and non-LC regions (IL_T_1, shared-3 callers for both types). Bold marks the higher value in each pair. [†]Small cells: $n = 50$, $n = 13$ and $n = 32$; the $n = 13$ cell is excluded from the tests. Non-LC regions have higher agreement in every sizable cell.

Denominator	SNV	Indel
TRUTH	0.32–0.72	0.14–0.52
UNION	0.12–0.16	0.18–0.31
UNION-HC	0.02–0.06	0.04–0.06

Table 5: Adjusted odds ratio of unanimity for LC vs. non-LC sites (range over the five D1 runs). Models adjust for VAF bin, depth, low mappability and segmental duplications. All $p < 10^{-10}$. An odds ratio below 1 means LC sites are less often unanimous.

The penalty survives adjustment and varies with VAF. After adjusting for depth, low mappability and segmental duplications, the LC odds ratio for unanimity is 0.32–0.72 (SNVs) and 0.14–0.52 (indels) on truth variants, and 0.02–0.06 within HC regions of the union (table 5). The VAF $\times$ LC interaction is significant in 28 of 30 native models (LRT $p < 0.05$). The LC penalty is thus present at every measurable VAF, but it is not constant. It is largest in the 10–20% and $\geq 20\%$ bins.

4.3 Indels disagree more at high VAF; at low VAF it depends on the callers

At $\geq 20\%$ VAF, indels are always worse. Table 6 shows that indel disagreement exceeds SNV disagreement at $\geq 20\%$ VAF in every run, experiment and caller panel, by 0.06 to 0.52.

With native panels, the ranking reverses below 20% VAF. On truth variants in IL_T_1, indels disagree *less* than SNVs by 0.23, 0.59 and 0.28 in the three lower bins. D1B, D2 at 100% purity and MC3 show the same reversal. H3 is therefore rated mixed in 33 D1 contexts and all six D1B contexts.

The reversal comes from SNV-only weak callers. Figure 5 shows per-caller recall on truth variants. At 5–10% VAF, SNV recall ranges from 0.05 (SomaticSniper) to 0.82 (MuTect2). Unanimity cannot exceed the recall of the weakest caller, and SomaticSniper is only in the SNV panel. In a leave-one-caller-out analysis, dropping SomaticSniper raises truth SNV unanimity at 5–10% from 0.038 to 0.167. Dropping any other caller changes it by less than 0.01. For indels, dropping TNscope helps most.

With shared panels, indels are worse in most bins. With the same three callers for both types, indels disagree more at 10–20% and $\geq 20\%$ VAF. This holds on truth variants (+0.08, +0.08), on the union (+0.10, +0.34) and within HC regions (+0.18, +0.52). Below 10% VAF the differences are within ± 0.01 , because both types sit near the floor. The dilution series, which uses a shared panel and a caller-independent expected VAF, shows indels worse in every bin: +0.12 to +0.56 at 20% purity and +0.07 to +0.20 pooled over purities, with disjoint CIs. The pooled shared-panel regression gives an indel odds ratio of 0.69–0.97 on truth variants and 0.37–0.60 within HC regions of the union. The indel $\times$ VAF interaction is significant in 14 of 15 models.

Which cell disagrees most? With shared panels, indels in LC regions have the highest disagreement at $\geq 20\%$ VAF in all five runs, both on truth variants and within HC regions. With native panels, LC

Context	<5%	5–10%	10–20%	≥20%
TRUTH, native callers (5 vs. 3)	−0.23 [−0.32, −0.16]	−0.59	−0.28	+0.06 [+0.04, +0.09]
TRUTH, shared-3	−0.00 (ns)	−0.01 (ns)	+0.08	+0.08
UNION, shared-3	−0.00	+0.01	+0.10	+0.34
UNION-HC, shared-3	−0.00 (ns)	+0.01	+0.18	+0.52
D2 TRUTH, shared-4, purity 20%	+0.12	+0.36	+0.56	—
D2 TRUTH, shared-4, pooled	+0.07	+0.16	+0.19	+0.20
D3 MC3, native (5 vs. 3)	—	−0.21	−0.36	+0.20

Table 6: Within-bin difference in disagreement, indel minus SNV. Positive values support H3. Rows without a label are D1 IL_T_1. Brackets give Newcombe 95% CIs. “−0.00” marks a negative difference that rounds to zero, where agreement is near zero for both types; ns means not significant. Bold marks positive differences of at least 0.05. With native panels the sign reverses below 20% VAF; with shared panels it does not.

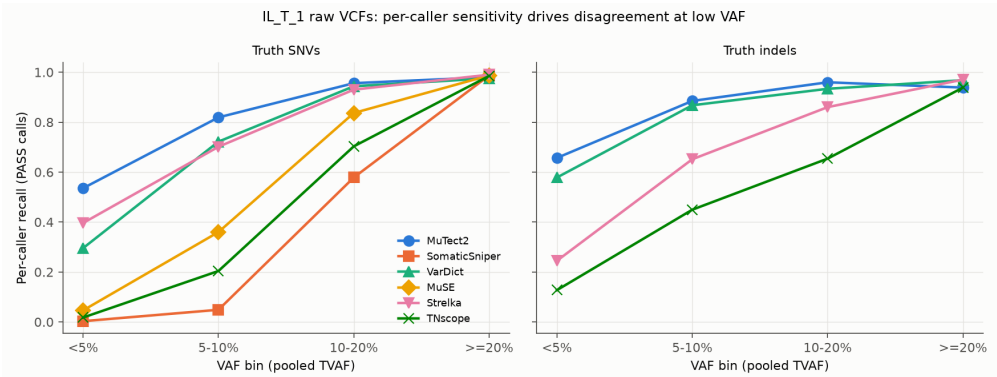


Figure 5: Per-caller recall (PASS calls) on truth variants in the IL_T_1 raw VCFs (D1B). (Left) SNVs. (Right) Indels. Callers converge above 0.9 at ≥20% VAF but spread widely below it. SomaticSniper (SNVs) and TNScope (both types) are the least sensitive at low VAF and so cap unanimity.

SNVs rank highest below 20% VAF. “Indels in repeats are the least agreed-upon variants” is therefore true for clonal variants but not in general.

4.4 Comparison with published numbers

Our results agree with prior work where the settings overlap. For the MuTect2–Strelka pair, the closest analogue to Chen et al. [2020], the union Jaccard in IL_T_1 is 0.82 at ≥20% VAF and 0.50 at 5–10%. With five or six callers, unanimity at 5–10% falls to at most 5%, because agreement shrinks multiplicatively as callers are added. Our union unanimity at ≥20% VAF (0.29–0.56 for SNVs) matches the 46% of Goode et al. [2013]. Over all truth SNVs, regardless of VAF, 69–77% are unanimous in the standard-depth runs (52% at 380×), close to the 82% reported by Wang et al. [2013]. We also reproduce the poor calling of repeat indels [Alioto et al., 2015] and the low-VAF failure of SomaticSniper [Wang et al., 2020].

5 Discussion

The denominator decides H1. Callers agree on real clonal mutations, but they do not agree on what they call. Between 63% and 70% of all calls come from a single caller, and most of these are low-VAF or LC artefacts; 98% of union SNVs below 5% VAF are singletons. A statement such as “callers agree on more than 80% of variants above 20% VAF” is true only for validated variants. This reconciles the apparently conflicting 46% [Goode et al., 2013] and 82% [Wang et al., 2013] in the literature. We recommend that every agreement number state its denominator.

Low-VAF disagreement is a sensitivity problem. On real variants below 10% VAF, callers disagree mainly because some of them miss the variant, not because of calling noise. On truth SNVs at 5–10% VAF, recall ranges from 0.05 to 0.82 (figure 5), and unanimity is bounded by the weakest caller. MC3 shows the same pattern independently, since its SomaticSniper also has zero recall below 10%. A unanimous-vote filter thus behaves like the least sensitive caller in the panel. For subclonal work, majority voting or caller-specific weighting is the better choice. For example, majority agreement on IL_T_1 truth SNVs is 0.647 at 5–10% VAF, compared with 0.044 for unanimity.

LC context is the most robust penalty. LC regions reduce unanimity for both variant types, in every dataset, after adjusting for mappability and segmental duplications, and with no reversed bin. The penalty is largest in the call union (odds ratio 0.02–0.16), because LC regions produce most private artefacts. It is also present among validated variants (odds ratio 0.14–0.72), so consensus filtering removes real repeat-region mutations as well. This matters for microsatellite-unstable tumors, where the relevant mutations are exactly repeat indels.

The indel penalty is real but depends on the caller panel. When SNVs and indels are compared with their native panels, the SNV panel includes SomaticSniper, which is weak at low VAF, and the comparison reverses below 20% VAF. Two further factors may affect the comparison. First, pooled TVAF may underestimate indel VAF because of reference bias in alignment, which would place some indels in lower bins where they look well agreed. Second, the truth set may favour well-supported low-VA indels. The shared-panel and dilution analyses avoid the panel confound, and they show indels worse in most bins. Researchers comparing agreement across variant types should therefore use matched caller panels.

Surprises. Two results were unexpected. First, deeper sequencing lowered agreement. At 380 $\times$, MuTect2 and MuSE missed 28–32% of true high-VA SNVs, and high-VA unanimity fell to 0.67. Filters tuned for standard depth may not transfer to very deep data. Second, Fleiss' κ is negative for union indels (–0.12 overall, –0.31 within HC regions). Indel callers agree less than chance given their marginal call rates, largely because VarDict makes many private indel calls.

Limitations.

- *Truth-set bias.* The SEQC2 truth set was built partly from these callers and excludes difficult regions. Agreement on the TRUTH denominator is therefore optimistic, and some LC truth cells are small (e.g., 13 LC indels below 5% VAF). The UNION denominator avoids this bias but mixes true and false calls.
- *One tumor with a truth set.* HCC1395 is a near-clonal cell line, so low-VA truth variants are scarce outside the dilution series. MC3 extends the results to other tumors, but it lacks a truth set and includes only variants with at least two callers, which makes its agreement an upper bound. The meaning of the * suffix in CENTERS is not fully documented. We count starred callers as calls; excluding them drives indel unanimity to zero, because almost all indel records contain a starred caller.
- *Fixed, released caller outputs.* We could not re-run callers, so modern tools such as DeepSomatic, and Strelka2 in MC3, are missing. SEQC2 has no SomaticSniper or MuSE indel calls, so native panels differ in size between types.
- *VA estimates and matching.* Union VA comes from different sources in D1 and D2. Exact matching of indels can create spurious disagreement when callers represent the same event differently. Normalisation mitigates this: in D2 at 100% purity, 95.5% of truth indels are recovered by at least one caller.
- *Statistics.* With up to 10⁶ sites, trivial differences reach significance, so we emphasise effect sizes. We model within-tumor dependence only in MC3. Our LC definition is a single GIAB stratum with 5 bp slop; we did not analyse homopolymer and tandem-repeat sub-strata separately.

Broader implications. Consensus calling is a precision tool. It works well for clonal variants in unique sequence, but it quietly removes subclonal mutations and repeat-region variants, which are biologically important in resistance and microsatellite instability. Studies that report “high-confidence” consensus mutations should expect these classes to be under-represented. This bias can distort estimates of tumor heterogeneity and of mutational signatures dominated by indels.

6 Conclusion

We tested whether somatic variant callers agree on the same tumor, stratifying jointly by VAF, variant type and LC context. We used five SEQC2 runs, a six-caller replicate, a purity dilution series and 10,074 TCGA exomes. On true variants, callers agree on more than 80% at $\text{VAF} \geq 20\%$ in four of five runs and in MC3 SNVs. They agree on fewer than 50%, usually fewer than 5%, below 5% VAF. The dilution series shows that VAF itself causes this drop. On everything the callers call, however, high-VAF unanimity is only 29–56%. LC regions disagree more for both SNVs and indels at every VAF where agreement is above zero, and the gap varies with VAF. Indels disagree more than SNVs at high VAF and with matched callers. At low VAF the ranking depends on the caller panel, so “indels are worse regardless of VAF” does not hold in general.

The key takeaway is that **unanimous consensus is a precise filter for clonal variants in unique sequence, but it removes nearly all true subclonal variants and much of the true signal in repeats**. We recommend reporting the denominator of any agreement estimate, avoiding all-caller consensus for subclonal and repeat-region variants, and comparing variant types only with matched caller panels.

Future work should re-run modern callers such as DeepSomatic, Strelka2 and recent Mutect2 releases on the SEQC2 purity and depth series. It should also analyse finer repeat strata by homopolymer length and repeat period, match indels with haplotype-aware tools such as vcfeval or som.py, and replicate the analysis on a second tumor with a somatic truth set, such as COLO829.

References

- Tyler S. Alioto, Ivo Buchhalter, Sophia Derdak, Barbara Hutter, Matthew D. Eldridge, et al. A comprehensive assessment of somatic mutation detection in cancer using whole-genome sequencing. *Nature Communications*, 6:10001, 2015.
- David Benjamin, Takuto Sato, Kristian Cibulskis, Gad Getz, Chip Stewart, and Lee Lichtenstein. Calling somatic SNVs and indels with Mutect2. *bioRxiv*, page 861054, 2019.
- Lei Cai, Wei Yuan, Zhou Zhang, Lin He, and Kuo-Chen Chou. In-depth comparison of somatic point mutation callers based on different tumor next-generation sequencing depth data. *Scientific Reports*, 6:36540, 2016.
- Zixi Chen, Yuchen Yuan, Xiaoshi Chen, Jiayun Chen, Shudai Lin, Xingsong Li, and Hongli Du. Systematic comparison of somatic variant calling performance among different sequencing depth and mutation frequency. *Scientific Reports*, 10:3501, 2020.
- Kristian Cibulskis, Michael S. Lawrence, Scott L. Carter, Andrey Sivachenko, David Jaffe, Carrie Sougnez, Stacey Gabriel, Matthew Meyerson, Eric S. Lander, and Gad Getz. Sensitive detection of somatic point mutations in impure and heterogeneous cancer samples. *Nature Biotechnology*, 31(3):213–219, 2013.
- Louise de Schaetzen van Brien, Maarten Larmuseau, Kim Van der Eecken, Frederic De Ryck, Pauline Robbe, Anna Schuh, Jan Fostier, Piet Ost, and Kathleen Marchal. Comparative analysis of somatic variant calling on matched FF and FFPE WGS samples. *BMC Medical Genomics*, 13, 2020.
- Nathan Dwarshuis, Divya Kalra, Jennifer McDaniel, et al. The GIAB genomic stratifications resource for human reference genomes. *Nature Communications*, 15, 2024.
- Kyle Ellrott, Matthew H. Bailey, Gordon Saksena, Kyle R. Covington, Cyriac Kandoth, Chip Stewart, et al. Scalable open science approach for mutation calling of tumor exomes using multiple genomic pipelines. *Cell Systems*, 6(3):271–281, 2018.
- Adam D. Ewing, Kathleen E. Houlahan, Yin Hu, Kyle Ellrott, Cristian Caloian, et al. Combining tumor genome simulation with crowdsourcing to benchmark somatic single-nucleotide-variant detection. *Nature Methods*, 12(7):623–630, 2015.

- Yu Fan, Liu Xi, Daniel S. T. Hughes, et al. MuSE: accounting for tumor heterogeneity using a sample-specific error model improves sensitivity and specificity in mutation calling from sequencing data. *Genome Biology*, 17:178, 2016.
- Li Tai Fang, Pegah Tootoonchi Afshar, Aparna Chhibber, Marghoob Mohiyuddin, Yu Fan, John C. Mu, et al. An ensemble approach to accurately detect somatic mutations using SomaticSeq. *Genome Biology*, 16:197, 2015.
- Li Tai Fang, Bin Zhu, Yongmei Zhao, Wanqiu Chen, Zhaowei Yang, et al. Establishing community reference samples, data and call sets for benchmarking cancer mutation detection using whole-genome sequencing. *Nature Biotechnology*, 39:1151–1160, 2021.
- David L. Goode, Sally M. Hunter, Maria A. Doyle, Tao Ma, Simone M. Rowley, David Choong, Georgina L. Ryland, and Ian G. Campbell. A simple consensus approach improves somatic mutation prediction accuracy. *Genome Medicine*, 5:90, 2013.
- A. Guille et al. A benchmarking study of individual somatic variant callers and voting-based ensembles for whole-exome sequencing. *Briefings in Bioinformatics*, 26(1):bbae697, 2025.
- Wendell Jones, Binsheng Gong, Natalia Novoradovskaya, Dan Li, Rebecca Kusko, et al. A verified genomic reference sample for assessing performance of cancer panels detecting small variants of low allele frequency. *Genome Biology*, 22:111, 2021.
- Anne Bruun Krøigård, Mads Thomassen, Anne-Vibeke Lænkholm, Torben A. Kruse, and Martin Jakob Larsen. Evaluation of nine somatic variant callers for detection of somatic mutations in exome and targeted deep sequencing data. *PLoS One*, 11(3):e0151664, 2016.
- Sayed Mohammad Ebrahim Sahraeian, Ruolin Liu, Bayo Lau, Karl Podesta, Marghoob Mohiyuddin, and Hugo Y. K. Lam. Deep convolutional neural networks for accurate somatic mutation detection. *Nature Communications*, 10:1041, 2019.
- Aldo Sergi, Luca Beltrame, Sergio Marchini, and Marco Masseroli. Integrated approach to generate artificial samples with low tumor fraction for somatic variant calling benchmarking. *BMC Bioinformatics*, 25, 2024.
- Mingyi Wang, Wen Luo, Kristine Jones, Xiaopeng Bian, Russell Williams, Herbert Higson, Dongjing Wu, Belynda Hicks, Meredith Yeager, and Bin Zhu. SomaticCombiner: improving the performance of somatic variant calling based on evaluation tests and a consensus approach. *Scientific Reports*, 10:12898, 2020.
- Qingguo Wang, Peilin Jia, Fei Li, Haiquan Chen, Hongbin Ji, Donald Hucks, Kimberly Brown Dahlman, William Pao, and Zhongming Zhao. Detecting somatic point mutations in cancer genome sequencing data: a comparison of mutation callers. *Genome Medicine*, 5:91, 2013.
- Wenming Xiao, Luyao Ren, Zhong Chen, Li Tai Fang, Yongmei Zhao, et al. Toward best practice in cancer mutation detection with whole-genome and whole-exome sequencing. *Nature Biotechnology*, 39:1141–1150, 2021.
- Yongmei Zhao, Li Tai Fang, Tsai-wei Shen, Sulbha Choudhari, Keyur Talsania, et al. Whole genome and exome sequencing reference datasets from a multi-center and cross-platform benchmark study. *Scientific Data*, 8:296, 2021.

A Supplementary results

A.1 Hypothesis tally

Table 7 counts the configurations in which H1 and H2 are supported. A configuration is a run, denominator, flag definition and caller panel. Table 8 counts the H3 and H4 verdicts across contexts. Here a context is a run or purity, denominator and caller panel.

Experiment	Hypothesis	SNV	Indel
D1 SEQC2 ensemble	H1	12/40	12/40
	H2	40/40	37/40
D1B raw VCFs (6/4 callers)	H1	2/6	2/6
	H2	6/6	6/6
D2 purity series	H1	11/32	0/30
	H2	36/36	36/36
D3 TCGA MC3	H1	2/2	0/2
	H2	2/2	–

Table 7: Configurations in which H1 or H2 is supported (Holm-adjusted one-sided binomial test). In D1, H1 is supported only on the TRUTH denominator and fails on every UNION configuration.

Experiment	Test	Supported (all)	Partial	Mixed	Reversed	Not sig.
D1	H3 indel > SNV	1	5	33	6	0
	H4 LC > non-LC	39	50	1	0	0
D1B	H3	0	0	6	0	0
	H4	4	8	0	0	0
D2	H3	13	2	17	4	0
	H4	24	44	0	0	4
D3	H3	0	1	1	0	0
	H4	0	3	0	0	1

Table 8: H3 and H4 verdicts across contexts. H4 is never reversed or mixed, except in one D1 context. H3 is often mixed, because the native-panel comparisons reverse below 20% VAF.

A.2 Sanity checks

- *Truth matching.* Between 93.6% and 99.99% of truth SNVs appear in the ensemble tables. The numbers absent are 2,506 in FD_T_1, 1,276 in IL_T_1 and 5 in NS_380X. Between 93% and 99.8% of truth indels are present.
- *Caller recall.* Every caller has recall of at least 0.9 at $\geq 20\%$ VAF, except MuTect2 and MuSE in the 380 $\times$ run.
- *Zero SNV unanimity below 10% VAF.* We traced this to SomaticSniper. At 100% purity, among sites with median VAF below 10%, SomaticSniper shared only 56 sites with TNscope, 18 with Lancet and one each with MuTect2 and Strelka. The zero is not a parsing error.
- *Robustness to flags and VAF source.* The lenient flag definition and the observed-VAF truth denominator give the same qualitative conclusions as the primary analysis.
- *Reproducibility.* Re-running the D3 pipeline from scratch produced byte-identical summary and bootstrap tables. All other steps are deterministic.

A.3 Software

We used Python 3.12.8, pandas 3.0.5, numpy 2.5.3, scipy 1.18.1, statsmodels 0.15.0, pysam 0.24.1, pyfaidx 0.9.0.4, matplotlib 3.11.2 and bedtools 2.31.0, on an AMD EPYC 7302 CPU (32 threads, no GPU). SomaticSeq source code defined the flag semantics of the ensemble tables. The data sources are the SEQC2 somatic reference (NCBI FTP), the TCGA MC3 v0.2.8 public MAF (GDC), GIAB stratifications v3.3, and UCSC hg38/hg19.